Sintrix Powder for IM/IV injection

[Ingredient]

Each vial contains：
Ceftriaxone (as Disodium)…………500 mg (potency, for pack of 500 mg)
Ceftriaxone (as Disodium)…………1.0 g (potency, for pack of 1.0 g)

[Description]

It is a white to yellowish orange crystalline powder in a seal vial. The color of the vial is colorless.

Reconstitution solution:

- The reconstitution solution is not included in this product.
- The use of freshly prepared solutions is recommended.
- After reconstitution, constituted solutions may vary from light yellow to amber, depending on length of storage, diluent, and concentration, without affecting their potency.
- Preparation of reconstitution solution (Method of Administration, see [Recommended Dose])

[Pharmacodynamics]

Microbiology

The bactericidal activity of ceftriaxone results from inhibition of cell wall synthesis. Ceftriaxone has a high degree of stability in the presence of beta-lactamases, both penicillinases and cephalosporinases of gram-negative and positive bacteria.

Gram-Negative Aerobes:
Acinetobacter calcoaceticus
Enterobacter aerogenes
Enterobacter cloacae
Escherichia coli
Haemophilus influenzae (including ampicillin-resistant and beta-lactamase producing strains)
Haemophilus parainfluenzae
Klebsiella oxytoca
Klebsiella pneumoniae
Moraxella catarrhalis (including beta-lactamase producing strains)
Morganella morganii
Neisseria gonorrhoeae (including penicillinase - and nonpenicillinase-producing strains)
Neisseria meningitidis
Proteus mirabilis
Proteus vulgaris
Serratia marcescens
Ceftriaxone is also active against many strains of Pseudomonas aeruginosa.

NOTE: Many strains of the above organisms that are multiply resistant to other antibiotics, eg, penicillins, cephalosporins and aminoglycosides, are susceptible to ceftriaxone.

Gram-Positive Aerobes:
Staphylococcus aureus (including penicillinase-producing strains)
Staphylococcus epidermidis
Streptococcus pneumoniae
Streptococcus pyogenes
Viridans group streptococci
NOTE: Methicillin-resistant staphylococci are resistant to cephalosporins, including ceftriaxone. Most strains of Group D streptococci and enterococci, e.g., Enterococcus (Streptococcus) faecalis, are resistant.

Anaerobes:
Bacteroides fragilis
Clostridium species
Peptostreptococcus species
NOTE: Most strains of C. difficile are resistant.
Ceftriaxone also demonstrates in vitro activity against most strains of the following microorganisms, although the clinical significance is unknown:

Gram-Negative Aerobes:
Citrobacter diversus
Citrobacter freundii
Providencia species (including Providencia rettgeri)
Salmonella species (including S. typhi)
Shigella species
Gram-Positive Aerobes:
Streptococcus agalactiae

ANAEROBES:
Bacteroides bivius
Bacteroides melaninogenicus

[Pharmacokinetics]

Ceftriaxone has extremely long plasma elimination half-life of approximately 8 hours. Like other cephalosporins, it is widely distributed through the body including gallbladder, lungs, bone, bile, CSF (diffuses into the CSF at higher concentrations when the meninges are inflamed) ; crosses placenta, reaches amniotic fluid and milk. Absorption : IM : Well absorbed. The bioavailability of ceftriaxone administered IM is approximately 100% and time to peak serum concentration is within 1-2 hours (IM) Potein binding : 85% to 90% Ceftriaxone is reversibly bound to albumin, and the binding decrease with the increase in the concentration. Elimination : Excreted unchanged in urine (33% to 65%) by glomerular filtration and in feces. Penetration into Particular Tissues : Ceftriaxone penetrates the inflamed meninges of neonates, infants and children.

[Indications]

Treatment of lower respiratory tract infections, skin and skin structure infections, bone and joint infections, intra-abdominal and urinary tract infections, sepsis and meningitis due to susceptible organism; perioperative prophylaxis of infections.

[Recommended Dose]

Standard dosage

Adults and children over 12 years

The usual dosage is 1–2 g of ceftriaxone once daily (every 24 hours). In severe cases or in infections caused by moderately sensitive organisms, the dosage may be raised to 4 g, once daily.

Duration of therapy

The duration of therapy varies according to the course of the disease. As with antibiotic therapy in general, administration of ceftriaxone should be continued for a minimum of 48-72 hours after the patient has become afebrile or evidence of bacterial eradication has been obtained.

Combination therapy

Synergy between ceftriaxone and aminoglycosides has been demonstrated with many gram-negative bacteria under experimental conditions. Although enhanced activity of such combinations is not always predictable, it should be considered in severe, life threatening infections due to microorganisms such as Pseudomonas aeruginosa. Due to chemical incompatibility between ceftriaxone and aminoglycosides, the two drugs must be administered separately at the recommended dosages. Chemical incompatibility with ceftriaxone has also been observed with i.v. administration of amsacrine, vancomycin and fluconazole.

Method of administration

As a general rule the solutions should be used immediately after preparation.

Intramuscular injection. For i.m. injection, ceftriaxone 500 mg is dissolved in 2 ml, and ceftriaxone 1 g in 3.5 ml, of 1% lidocaine hydrochloride solution and injected well within the body of a relatively large muscle. It is recommended that not more than 1 g be injected at one site.

The lidocaine solution should never be administered intravenously.

Intravenous injection. For i.v. injection, ceftriaxone 500 mg is dissolved in 5 ml, and ceftriaxone 1 g in 10 ml, sterile water for injections. The intravenous administration should be given over 2–4 minutes.

Do not use diluents containing calcium, such as Ringer's solution or Hartmann's solution, to reconstitute ceftriaxone vials or to further dilute a reconstituted vial for i.v. administration because a precipitate can form. Precipitation of ceftriaxone-calcium can also occur when ceftriaxone is mixed with calcium-containing solutions in the same i.v. administration line. Ceftriaxone must not be administered simultaneously with calcium-containing i.v. solutions, including continuous calcium-containing infusions such as parenteral nutrition via a Y-site. However, in patients other than neonates, ceftriaxone and calcium-containing solutions may be administered sequentially of one another if the infusion lines are thoroughly flushed between infusions with a compatible fluid.

There have been no reports of an interaction between ceftriaxone and oral calcium-containing products or interaction between intramuscular ceftriaxone and calcium-containing products (i.v. or oral).

Special Dosage Instructions

Patients with hepatic impairment

In patients with liver damage, there is no need for the dosage to be reduced provided renal function is not impaired.

Patients with renal impairment

In patients with impaired renal function, there is no need to reduce the dosage of ceftriaxone provided hepatic function is not impaired. Only in cases of preterminal renal failure (creatinine clearance <10 ml/min) should the ceftriaxone dosage not exceed 2 g daily. Ceftriaxone is not removed by peritoneal- or hemodialysis. In patients undergoing dialysis no additional supplementary dosing is required following the dialysis.

Patients with severe renal and hepatic impairment

In patients with both severe renal and hepatic dysfunction, clinical monitoring for safety and efficacy is advised.

Elderly

The dosages recommended for adults require no modification in elderly patients, provided there is no severe renal and hepatic impairment.

Children

Neonates, infants and children up to 12 years

The following dosage schedules are recommended for once daily administration:

Neonates (up to 14 days): 20–50 mg/kg bodyweight once daily. The daily dose should not exceed 50 mg/kg. Ceftriaxone is contraindicated in premature neonates up to a postmenstrual age of 41 weeks (gestational age + chronological age).

Ceftriaxone is contraindicated in neonates (≤28 days) if they require (or are expected to require) treatment with calcium-containing i.v. solutions, including continuous calcium-containing infusions such as parenteral nutrition, because of the risk of precipitation of ceftriaxone-calcium.

For neonates, infants, and children (15 days to 12 years): 20–80 mg/kg once daily.

For children with bodyweights of 50 kg or more, the usual adult dosage should be used.

Intravenous doses of ≥50 mg/kg bodyweight, in infants and children up to 12 years of age, should be given by infusion over at least 30 minutes. In neonates, intravenous doses should be given over 60 minutes to reduce the potential risk of bilirubin encephalopathy.

Meningitis

In bacterial meningitis in infants and children, treatment begins with doses of 100 mg/kg (up to a maximum of 4 g) once daily. As soon as the causative organism has been identified and its sensitivity determined, the dosage can be reduced accordingly. The following duration of therapy has shown to be effective:

Neisseria meningitidis	4 days
Haemophilus influenzae	6 days
Streptococcus pneumoniae	7 days

Lyme borreliosis

50 mg/kg to a maximum of 2 g in children and adults, once daily for 14 days.

Perioperative prophylaxis

A single dose of 1-2 g depending on the risk of infection 30-90 minutes prior to surgery. In colorectal surgery, administration of ceftriaxone with or without a 5-nitroimidazole, e.g. ornidazole has been proven effective.

[Administration]

IM/IV

[Contraindications]

The most common untoward reactions are limited essentially to gastrointestinal disturbances and hypersensitivity phenomena.

Ceftriaxone is contraindicated in neonates (<28 days of age) if they require (or are expected to require) treatment with calcium-containing intravenous solutions, including calcium-containing infusions such as parenteral nutrition, because of the risk of precipitation of ceftriaxone-calcium.

[Warning and Precautions]

Although transient elevations of BUN and serum creatinine have been observed, at the recommended dosages, the nephrotoxic potential of Ceftriaxone for Injection is similar to that of other cephalosporins. Ceftriaxone is contraindicated in patients who have had a previous hypersensitivity reaction to any cephalosporin. It is also contraindicated in patients who have had a previous immediate and/or any severe hypersensitivity reaction to any penicillin or to any other beta-lactam drug. Ceftriaxone should be given with caution to patients who have other allergic diatheses. Antibiotic-associated diarrhoea, colitis and pseudomembranous colitis have all been reported with the use of ceftriaxone. These diagnoses should be considered in any patient who develops diarrhoea during or shortly after treatment. Ceftriaxone should be discontinued if severe and/or bloody diarrhoea occurs during treatment and appropriate therapy instituted. Ceftriaxone should be used with caution in individuals with a previous history of gastro-intestinal disease, particularly colitis. In vivo and in vitro studies have shown that ceftriaxone, like some other cephalosporins, can displace bilirubin from serum albumin. Clinical data obtained in neonates have confirmed this finding. Therefore, patients with renal failure normally require no adjustment in dosage when usual doses of Ceftriaxone for Injection are administered, but concentrations of drug in the serum should be monitored periodically. If evidence of accumulation exists, dosage should be decreased accordingly. Dosage adjustments are not necessary in patients with hepatic dysfunction. Ceftriaxone must not be mixed or administered simultaneously with calcium contaning solutions or products, even via different infusion lines. Calcium containing solutions or products must not be administered within 48 hours of last administration of ceftriaxone. Cases of fatal reactions with calcium-ceftriaxone precipitates in lungs and kidneys in both term and premature neonates have been described. In some cases the infusion lines and times of administration and calcium-containing solutions differed.

In patients other than neonates, Ceftriaxone and calcium-containing solutions may be administered sequentially to one another if the infusion lines are thoroughly flushed between infusions with a compatible fluid.

Diluents containing calcium, such as Ringer’s solution or Hartmann’s solution, are not to be used to reconstitute Ceftriaxone vials or to further dilute a reconstituted vial for intravenous administration because a precipitate can form. Ceftriaxone must not be administered simultaneously with calcium-containing intravenous solutions, including continuous calcium-containing infusions such as parenteral nutrition via a Y-site, because precipitation of ceftriaxone-calcium can occur.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with bata-lactams. Before initiating therapy with Sintrix, carefully inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Sintrix must be discontinued immediately and appropriate alternative instituted.

[Interaction with other medicaments]

No impairment of renal function has so far been observed after concurrent administration of large doses of ceftriaxone and potent diuretics (e.g. furosemide). There is no evidence that ceftriaxone increases renal toxicity of aminoglycosides. No effect similar to that of disulfiram has been demonstrated after ingestion of alcohol subsequent to the administration of ceftriaxone. Ceftriaxone does not contain an N-methylthiotetrazole moiety associated with possible ethanol intolerance and bleeding problems of certain other cephalosporins. The elimination of ceftriaxone is not altered by probenecid. In an in-vitro study antagonistic effects have been observed with the combination of chloramphenicol and ceftriaxone.

[Pregnancy and Lactation]

Usage in Pregnancy: There are no adequate and well-controlled studies in pregnant women; therefore, this drug should be used during pregnancy only if clearly needed.

Nursing Mothers: Low concentrations of Ceftriaxone are excreted in human milk. Caution should be exercised when Ceftriaxone is administered to a nursing woman.

[Side Effects]

(1) Gastrointestinal:diarrhea, vomiting, nausea and stomatitis.
(2) Dermatologic.
(3) Hematologic:eosinophilia, leukopenia, granulocytopenia.
(4) Dermatological:allergic dermatitis, pruritus, edema, urticaria.
(5) Other rare side effects:headache, dizziness, increase in liver enzymes & serum

[Symptoms and Treatment of Overdose]

In the case of overdosage, drug concentration would not be reduced by hemodialysis or peritoneal dialysis. There is no specific antidote. Treatment of overdosage should be symptomatic.

[Incompatibilities]
Solutions containing ceftriaxone should not be mixed with or added to solutions containing other agents except 1% Lidocaine Injection BP (for intramuscular injection only). In particular, ceftriaxone is not compatible with calcium-containing solutions such as Hartmann's solution and Ringer's solution. Based on literature reports, ceftriaxone is not compatible with amsacrine, vancomycin, fluconazole, aminoglycosides, pentamidine and labetalol.

[Package]

Pack of 500 mg : 500 mg (potency) per vial, 10 vials per paper box.
Pack of 1.0 g : 1.0 g (potency) per vial, 10 vials per paper box.

[Storage]

Store below 30℃ and protect from light.
Discard any unused portion after opening.

KEEP OUT OF REACH OF CHILDREN

[Shelf-life]

Three years.

Registration No.：Pack of 500 mg : MAL20152503AZ
Pack of 1.0 g : MAL20152504AZ

Manufacturer:
TAIWAN BIOTECH CO., LTD. KUAN YIN PLANT
No. 1, Kwo Chien 1st Road, Kuan Yin District, Taoyuan City,
Taiwan, R.O.C.

Importer and Product Registration Holder:
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